

TRAPPES, France

WMO: 071450

Lat: 48.7744N

Lon: 2.0097E

Elev: 571

StdP: 14.40

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.7	26.5	16.2	12.8	27.0	20.2	15.5	29.3	18.4	48.9	16.7	48.7	4.4	10	0.468

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.3	86.9	67.8	82.8	66.2	79.4	64.9	69.5	82.9	67.7	79.4	66.2	76.6	5.8	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.3	95.6	73.8	63.7	90.1	71.6	62.1	85.3	70.3	33.9	83.0	32.4	79.4	31.2	76.9	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	15.2	13.5	11.9	19.6	94.2	4.4	4.5	16.4	97.4	13.8	100.0	11.3	102.5	8.0	105.8	
(5)				18.6	73.1	4.5	1.7	15.4	74.4	12.7	75.4	10.2	76.3	7.0	77.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.7	39.6	40.5	45.7	51.2	57.3	63.4	66.8	66.1	60.9	54.0	45.7	40.6	(6)
(7)		DBStd	11.57	6.86	6.79	6.31	6.26	6.02	5.88	5.86	5.35	5.06	6.14	6.12	6.67	(7)
(8)		HDD50	1305	326	271	162	57	8	0	0	0	31	154	296		(8)
(9)		HDD65	4749	789	687	599	415	250	99	46	47	142	341	579	755	(9)
(10)		CDD50	2299	2	3	28	93	234	402	522	500	328	157	25	5	(10)
(11)		CDD65	271	0	0	0	1	11	52	103	82	21	1	0	0	(11)
(12)		CDH74	2359	0	0	0	19	96	437	913	719	166	9	0	0	(12)
(13)		CDH80	690	0	0	0	0	8	119	291	244	28	0	0	0	(13)
(14)	Wind	WSAvg	5.6	6.4	6.5	6.3	5.7	5.4	5.3	5.1	4.8	4.6	5.1	5.7	6.4	(14)
(15)	Precipitation	PrecAvg	26.5	2.3	1.9	2.2	1.9	2.4	2.1	2.4	2.0	2.1	2.4	2.4	2.3	(15)
(16)		PrecMax	33.6	4.7	4.2	4.3	4.4	4.6	6.3	6.0	4.5	5.0	6.5	3.9	4.6	(16)
(17)		PrecMin	16.3	0.5	0.3	0.1	0.1	0.3	0.1	0.6	0.1	0.3	0.3	0.4	0.2	(17)
(18)		PrecStd	4.4	1.1	1.1	1.1	1.1	1.2	1.2	1.3	1.1	1.3	1.7	0.9	1.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.6	59.9	68.0	76.7	80.6	88.5	92.7	94.2	84.0	74.6	62.3	56.8	(19)
(20)		MCWB	52.7	49.9	52.9	60.1	66.1	69.7	69.4	70.4	65.7	62.6	57.9	53.5		(20)
(21)		2%	DB	53.2	54.6	63.3	72.1	76.7	83.4	86.8	86.7	79.1	69.5	59.3	54.3	(21)
(22)		MCWB	50.8	48.7	52.0	57.3	62.0	67.0	67.6	67.4	64.2	61.3	56.0	51.7		(22)
(23)		5%	DB	51.1	52.2	59.5	66.7	72.9	78.8	82.7	81.6	74.5	65.5	56.7	52.3	(23)
(24)		MCWB	49.0	48.3	50.6	54.7	60.5	65.0	66.2	65.6	62.5	59.2	53.6	50.0		(24)
(25)		10%	DB	49.0	50.1	55.6	62.7	68.7	75.1	79.1	77.3	70.9	62.5	54.3	50.3	(25)
(26)		MCWB	46.9	46.8	49.4	52.8	58.5	63.4	65.1	63.6	60.6	57.6	51.7	48.1		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.6	52.3	55.6	62.0	67.3	72.3	71.3	71.6	67.8	64.5	59.3	54.2	(27)
(28)		MCDWB	55.1	55.9	62.4	74.1	78.5	84.3	86.3	87.8	78.9	70.7	61.1	55.9		(28)
(29)		2%	WB	51.2	50.4	53.6	58.6	64.4	68.7	69.3	68.9	65.5	61.8	56.3	52.2	(29)
(30)		MCDWB	52.6	53.2	60.1	69.2	73.5	79.4	83.4	82.2	75.0	67.1	58.6	53.9		(30)
(31)		5%	WB	49.2	48.8	52.0	55.9	61.7	66.5	67.6	67.0	63.8	60.2	53.9	50.4	(31)
(32)		MCDWB	50.8	51.5	57.5	64.8	69.9	75.9	79.7	78.3	71.7	64.7	56.2	52.2		(32)
(33)		10%	WB	47.1	47.1	50.2	53.7	59.5	64.5	66.0	65.2	62.2	58.5	51.9	48.4	(33)
(34)		MCDWB	48.8	49.8	55.3	60.9	66.8	73.5	76.8	74.8	69.1	62.0	54.1	50.2		(34)
(35)	Mean Daily Temperature Range	MDBR	8.3	10.8	13.9	16.1	16.1	16.6	17.3	17.5	16.6	13.0	9.6	8.7		(35)
(36)		5% DB	MCDBR	9.1	14.0	20.2	22.5	21.8	22.6	23.3	25.0	22.4	16.3	11.2	9.2	(36)
(37)		MCWBR	8.1	9.7	11.1	10.9	10.2	10.1	8.7	9.7	9.6	9.3	8.4	8.0		(37)
(38)		5% WB	MCDWB	8.8	11.8	16.1	20.3	19.2	19.4	20.9	21.1	18.6	14.5	10.2	8.7	(38)
(39)		MCWBR	8.2	9.3	10.1	10.4	10.0	9.9	8.5	9.4	9.1	9.0	8.3	8.1		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.330	0.351	0.385	0.403	0.409	0.411	0.410	0.400	0.374	0.368	0.340	0.324		(40)
(41)		taud	2.394	2.350	2.249	2.217	2.235	2.266	2.265	2.314	2.375	2.381	2.401	2.367		(41)
(42)		Ebn at Noon	229	251	260	267	270	270	268	266	262	241	222	213		(42)
(43)		Edh at Noon	22	28	36	41	42	41	41	37	32	27	21	19		(43)
(44)	All-Sky Solar Radiation	RadAvg	291	539	910	1382	1628	1817	1722	1496	1156	681	352	246		(44)
(45)		RadStd	32	75	113	156	153	173	138	142	92	56	34	36		(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page